

Bold Sky: A Community-Based Intimate Partner Violence Prevention Digital Game

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Abstract

Intimate partner violence (IPV) is a pervasive global issue, affecting one in three women worldwide, with particularly high rates of perpetration among men. Roots causes of IPV include harmful social and gender norms (SGNs). Despite existing prevention strategies, many focus on interventions after the violence has occurred, leaving a critical gap in primary prevention. My research seeks to fill this gap by developing “Bold Sky” a gender-transformative digital game aimed at preventing IPV perpetration among young men. Grounded in Human-Computer Interaction, public health, and social work principles, this research will use co-design and community-based methodologies to engage young men and IPV prevention experts in its development to support relevance to this target group and evaluation of the game to ensure effectiveness in shifting harmful SGNs. The aim for this research is to co-design and develop “Bold Sky” and launched for real-world impact.

CCS Concepts

• **Software and its engineering** → **Designing software; Interactive games**; • **Information systems** → **Collaborative and social computing systems and tools**; • **Applied computing** → **Law, social and behavioral sciences; Media arts; Interactive learning environments**.

Keywords

Digital Games, Serious Games, Violence Prevention, Intimate Partner Violence, Community-Based Research

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1 Introduction & Motivation

Intimate partner violence (IPV) is a critical, but preventable, public and global health issue that has significant impacts on affected individuals, families, communities and societies. The World Health

Organization (WHO) defines IPV as “behaviour by an intimate partner or ex-partner that causes physical, sexual or psychological harm, including physical aggression, sexual coercion, psychological abuse and controlling behaviours” [18] with an estimated one in three women having experienced physical and/or sexual IPV in their lifetimes [18]. These statistics do not account for other, less “visible” forms of IPV such as psychological abuse, harassment, and stalking [8].

Since the COVID-19 outbreak, there has been a global increase in requests for IPV assistance and a simultaneous decrease in capacity for support [24]. Global lockdown measures and economic pressures have led to as much as a 54% increase in reports of IPV [16]. IPV victims have lost income, and thus their financial independence, due to the pandemic response, and safe spaces like IPV shelters have reduced capacity to comply with public health measures [25]. Organizations have also had to rapidly adapt their service delivery methods [13, 24]. Many social service organizations and legal proceedings struggled to adapt to provide supports and services digitally, leading to lengthy delays and increased safety concerns for IPV victims [1, 14]. These stresses during the pandemic highlighted the flaws in post-violence intervention as well as the status quo delivery of in-person support, and underscore the pressing need for new kinds of supports. UN Women has issued a global call to stop this “Shadow Pandemic” [18].

1.1 Personal Motivation

As a registered social worker in Canada specializing in IPV and gender-based violence (GBV) sectors, I have seen first-hand the devastation and negative images that IPV & GBV causes. These negative impacts can include poor physical, mental, emotional, social, and economic well-being. Additionally, for children who have experienced IPV, these experiences can also negatively impact their developmental growth, mental health, and social connectedness. IPV can also turn deadly. In a 2023 report by the Canadian Femicide Observatory for Justice and Accountability, it was reported that over 850 women and girls were killed between 2018-2022 — an average of one woman or girl being killed every two days [9], with over 70% of female homicide victims were killed by their current or previous partners in 2020 [2].

Beyond the individual consequences of IPV, there are also societal costs to IPV. In 2019, an estimated \$7 billion dollars lost due to spousal violence in Canada, a subset of IPV [29]. These costs arise from the provision of medical, judicial, and social intervention, lost wages and education, pain and suffering to survivors of spousal violence and their families, negative impact on children, and losses to employers [29]. While Canada does provide funding and social

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service to support IPV victims (or punish IPV perpetrator), very few resources are funneled into primary prevention (preventing IPV incidents from occurring from the first place).

IPV is a pervasive and complex issue, necessitating multifaceted prevention and intervention strategies. With the rise of digital technology, there has been growing interest in how technological solutions can contribute to preventing IPV. This is an area that I am passionate about contributing to. I aim to bridge the fields of Human-Computer Interaction (HCI), Public Health, and Social Work to develop innovative, responsive technologies that address the root causes of IPV and promote gender-transformative practices. By merging these disciplines, I hope to create scalable IPV and GBV prevention tools that are both technologically advanced and grounded in trauma-informed, evidence-based approaches, ensuring that they are effective, accessible, and culturally sensitive.

2 Background & Related Works

In Canada, men between the ages of 25-34 constitute the largest group of IPV perpetrators [4]. Consequently, WHO has stressed the need for interventions targeted at young men before they reach this age [15, 23]. However, Canada has focused on post-violence resources and services [17]. Most of these post-violence interventions are resource-heavy as they rely on in-person engagement (i.e., police intervention or hospital visits) and do not address the root cause of the violence, including gender inequality and social norms that condone violence [18]. The WHO has emphasizes the importance of engaging in prevention programs, particularly for young men, to meaningfully address harmful gender and social norms to prevent IPV [18], such as using gender-transformative approaches (GTA).

2.1 Gender-Transformative Approaches

Research on prevention strategies for young men consistently emphasizes the role that harmful social and gender norms (SGNs) play in perpetuating IPV [10, 22, 27]. Traditional ideas of masculinity are associated with a higher likelihood of perpetrating and condoning violence towards women and girls [10, 22]. GTA – those that aim to shift harmful SGNs through (1) positive reinforcement of new gender equality norms; (2) negative reinforcement of harmful old norms; and (3) community role modeling [21] – have been successfully used around the world, including in India, South Africa, Rwanda, and the United States [20]. Almost all the GTA interventions involved in-person or face-to-face program delivery [20].

2.1.1 Key Problem/Gap. Current GTA interventions have a number of limitations. Most notably, they rely heavily on face-to-face program delivery [20], which can be resource-intensive (requiring trained facilitator(s)), expensive (accessible space, scheduling and materials needed to run the program), and difficult to scale effectively across diverse communities. Additionally, traditional in-person (or face-to-face) formats may not resonate with young men, who may feel uncomfortable by the idea of attending due to fears of judgment, shame, stigma or ambivalence around discussing sensitive topics like IPV and healthy masculinity [12, 26]. These barriers mean that many young men who could benefit from these programs may not participate, leaving a critical gap in prevention

efforts. These limitations create gaps in addressing IPV prevention – without the ability to scale-up due to resource limitation and the barriers for men to engage in-person programs – current interventions will continue to be unable to adequately address IPV.

2.2 The Role of Technology in IPV Prevention

Many digital technologies have been used in public health as tools for behavior change and education [5, 28]. Unlike traditional in-person programs, digital interventions can more easily scale-up by reducing logistical constraints and reluctance of attending face-to-face groups, particularly for sensitive topics like IPV [11, 26].

Despite these benefits, I have found in my scoping review that most existing digital IPV interventions focus on secondary and tertiary prevention, focusing on support after the violence has occurred. Few technologies have been designed and developed with the goal of primary prevention – preventing IPV from occurring in the first place. This gap highlights the need for digital tools that proactively challenge harmful SGNs. My scoping review has shown that interactive platforms like games can provide a more engaging space that supports IPV prevention.

2.3 Related Works

There have been a few games developed specifically for IPV prevention. These games are typically developed for youth and are created for educational and awareness-building purposes. For example:

2.3.1 Danielle. was a digital game developed by Ni3 with a focus on teen dating violence and prosocial behaviors in the United Kingdom [19]. Danielle's target audience are male and female 14-18 year olds in secondary school settings and conducted small focus groups involving 17-18 year olds. Danielle focused on awareness building and prosocial peer support, however it is inconclusive as to its ability to translate into behavioral changes.

2.3.2 PR:EPARe. or Positive Relationships: Eliminating Coercion and Pressure in Adolescent Relationships was a study and digital game-based approach to addressing dating violence in the United Kingdom [6]. PR:EPARe recruited a total of 505 female and male students, between the ages of 13-14 years old, split into experimental (digital game) and control groups (standard school relationship and sex education curriculum). The study found that the digital game resulted in greater awareness and effectiveness of addressing dating violence than compared to the control group.

2.3.3 Green Acres High. was a digital serious game released in multiple countries in Europe (England, Sweden, Germany and Belgium) and focused on dating violence education [3]. The study was focused on players' experiences of using serious games as an educational tool and recruited 13 participants between the ages 12-16 year olds. Researchers had the male and female participants play the digital game and participate in focus groups to get a deeper understanding of European perspectives and attitudes on dating violence. The study found that participants had a positive learning experience, increased understanding and increased confidence to identify and manage dating violence [3].

2.3.4 The Mystery of Pandora. was an immersive virtual reality (IVR) game developed to educate high school students and professionals about domestic violence against women (DVAW) [7]. The game was designed to educate and raise awareness about this form of violence by engaging players to play as a criminal investigating working on a murder case with a female victim. Felix et al. [7] found that IVR was a promising area of IPV prevention research as it provides players with a more realistic and engaging experience while exploring the nuances of (DVAW).

3 Research Objectives & Questions

My research will aim to address the current issues with IPV prevention (i.e., scalability, reliance on in-person/face-to-face program) while incorporating GTA and tailored specifically for young men (ages 18-25). A key part of the research will be to co-design a GTA IPV prevention game by engaging an interdisciplinary group of young men, experts in IPV prevention and community stakeholders with the goal to create an immersive, engaging and scale game that shift harmful SGNs. The game will be evaluated to assess the effectiveness of the game's ability to shift harmful SGNs to prevent IPV among self-identified young men.

3.1 Research Questions

This research project will explore the following research questions:

- RQ1:** What technological artifacts currently exist for intimate partner violence prevention and what are promising area for intimate partner violence prevention?
- RQ2:** How can digital games be designed to shift social and gender norms related to intimate partner violence?
- RQ3:** What elements of Gender-Transformative Approaches make intimate partner violence interventions effective for young men (ages 18-25)?
- RQ4:** How can co-design processes involving young men and community stakeholders enhance intimate partner violence prevention game design?

4 Research Design & Methodology

This research adopts a participatory co-design approach, where young men and IPV prevention experts will be invited to collaborate and develop a GTA digital game, which ensures that the game will be engaging, culturally relevant, and addresses the specific needs and experiences of young men. The project is structured in four phases:

- P1:** Scoping review on current technological tools to prevent IPV.
- P2:** Co-design and development sessions with young men and experts & playtesting game prototype.
- P3:** Evaluation of game outcomes.
- P4:** Real-world deployment & scale-up.

In Phase 1, I completed a scoping review that mapped out current technological tools that have been designed or developed to prevent IPV. This scoping review is currently under review for the CHI '25 conference. In Phase 2, young men and experts have been recruited to participate in co-design workshops to conceptualize the game's

mechanics and content as well as playtesting and refining the prototype based on user feedback. Phase 3 will evaluate the game's impact on shifting attitudes and behaviors related to IPV using a combination of quantitative and qualitative methods, ensuring a comprehensive understanding of its effectiveness. Adjustments to the game will be made to better support shift to SGNs to prevent IPV. Finally, in Phase 4, there will be real-world deployment of the digital game in Waterloo, Canada as well efforts to scale-up the deployment and adoption of the game across Canada and globally.

5 Rationale

The co-design methodology is essential to ensure the digital game is user-centered and culturally relevant, particularly for young men who may be reluctant to engage in traditional IPV prevention programs. Digital games provide an immersive and interactive platform, making them a promising tool for promoting behavior change. By incorporating GTA content and using HCI principles, my research project aims to create a scalable, accessible, and effective IPV prevention intervention.

5.1 Theoretical Frameworks:

This research is grounded in GTA and community-based frameworks, while incorporating co-design, and human-centered HCI principles. The GTA framework challenges harmful SGNs and promotes healthier, equitable relationships by actively engaging young men in rethinking masculinity and preventing IPV. Co-design methodology empowers the target population—young men—to participate in the creation process, ensuring that the digital game is culturally relevant, engaging, and responsive to their needs. Human-centered HCI principles guide the development of the game, focusing on creating an intuitive, accessible, and user-friendly experience that fosters empathy and encourages behavioral change

6 Completed & Future Work

6.1 Scoping Review

I completed a scoping review that examined current technologies designed to prevent IPV, using public health categories to organize the interventions (i.e., primary (focuses on stopping IPV before it starts by promoting healthy relationships and addressing harmful social norms), secondary (early intervention of IPV to reduce risks and prevent escalation), and tertiary prevention levels (providing treatment and support to IPV survivors and perpetrators to prevent further harm). One of the key findings was the limited focus on primary prevention, as most technologies were designed to intervene after IPV had already occurred. This highlighted a significant gap in preventing IPV before it happens.

The scoping review also found the importance of tailoring technologies to specific populations and contexts. Many existing technologies lacked the flexibility to account for diverse cultural, social, and gender-specific factors that influence IPV prevention needs across different groups. Another key finding was the lack of technologies designed specifically for young men, who are a critical demographic in efforts to prevent IPV.

One promising area identified in the review was the potential of digital games as IPV interventions. Digital games offer an interactive and engaging platform that can challenge harmful attitudes,

foster empathy, and encourage reflection on SGNs. This finding has informed my current research, which focuses on developing a GTA digital game for young men, aimed at serving as a primary prevention tool for IPV. The review provided essential insights into the gaps in existing technologies and pointed toward digital games as a promising solution to address these unmet needs.

6.2 Project Advisory Committee

I have made significant progress has been made in forming an interdisciplinary advisory committee to support Phases 2-4 of the project. I have engaged with a range of stakeholders, including young men, IPV prevention experts, industry game developers, HCI experts, social workers, product developers, and behavioral architects. Collaboration with research centers, such as The Games Institute, has also been established. This advisory committee will provide comprehensive support throughout the co-design process, ensuring the digital game is shaped by diverse perspectives, expertise, and practical insights to create an effective, culturally relevant tool for preventing IPV among young men.

6.3 Future Work

The next steps for the IPV prevention game project involve the following: first, the co-design sessions with young men and the advisory committee will continue, allowing for further refinement of the game's narrative, mechanics, and educational content. This will lead to the development of an early prototype, which will undergo iterative playtesting with the target audience to ensure it meets their needs and expectations. Additionally, securing ethics approval for the evaluation phase is a priority, as well as finalizing the evaluation framework, which will include both quantitative and qualitative methods to assess the game's impact on attitudes and behaviors. Planning for broader outreach and knowledge mobilization is also underway in preparation for the game's future scale-up.

7 Questions & Challenges

One of the most prominent question is to understand how effective the game will be in shifting harmful SGNs among young men. Additionally, there are also questions about the game's cultural adaptability, as the game will likely require adjustments to remain relevant across diverse regions, communities, and demographics. Another important consideration is how to engage young men, who may be reluctant to participate in traditional IPV prevention programs, and what strategies can be employed to ensure the game resonates with and is used by this target audience.

Some of the challenges I anticipate facing include recruiting and maintaining participation from young men and other stakeholders throughout the co-design and evaluation phases, and consistent feedback will be crucial for the game's success. Furthermore, balancing the game's educational content with engaging gameplay will be complex, as it is essential to motivate and immerse users while delivering meaningful GTA lessons. Technological limitations and accessibility also pose potential hurdles, especially in ensuring the game is available across various devices and platforms and meets the diverse needs of users (i.e., offering different language options

or usability features). Finally, measuring the game's long-term impact on behavior change will be challenging, as capturing shifts in SGNs over time requires careful planning and execution of the game evaluation.

8 Dissertation Status & Long-Term Goals

I am starting my fourth year as a PhD candidate in the School of Public Health Sciences at the University of Waterloo under the co-supervision of Dr. James R. Wallace and Dr. Cayley MacArthur. Canadian PhD programs typically span 4-5 years and I plan to defend my thesis in December 2025. I have not previously participated in a doctoral consortium. My long-term goal is to continue as an interdisciplinary researcher and educator in the HCI, Public Health and Social Work fields within academia. I also hope to continue to explore, develop and evaluate methods that create artifacts through community-based and participatory approaches to improve social justice and health equity for all.

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